



Unit 3 · Outcome 1 · Maintaining Biodiversity

Nobody planted the bush. The planet did.

KK05

VCE Environmental Science · Units 3 & 4 · Exam study summary

Digital Vector EnviroSci · single-topic mastery summary — clean explanations, comparison tables, visual aids, exam traps, command words and worked answers.



One idea, five forces

Everything in this dot point hangs off a single sentence — learn it and you can rebuild the rest.

◆ NOTE

The sentence to memorise Natural drivers change ecosystems over different time scales, and in doing so they create endemism, concentrate it into biodiversity hotspots, and drive both extinction and new diversification.

The five drivers VCAA names are volcanic eruptions, fire, droughts and floods linked to El Niño, plate tectonic movement, and evolution / natural selection. Two of these are slow giants working over millions of years; three are fast, repeating disturbances measured in years to centuries. The exam loves asking you to tell those time scales apart — so we'll build the whole module around them.

◆ ANALOGY

Analogy · the mixing desk Think of biodiversity as the sound coming out of a mixing desk. Evolution and plate tectonics are the slow master faders — nudged over millions of years. Fire, El Niño and eruptions are the fast effects pedals — stomped on every few years, sometimes wiping a track, sometimes adding a whole new layer. The track you finally hear is every fader and pedal combined.

"Different time scales" — the examiner's favourite phrase

Place each driver on the line. Short-term drivers reshuffle who's present; long-term drivers change what exists at all.

◆ NOTE

Why the distinction earns marks A short-term driver like an El Niño drought changes which species are present right now (population crashes, range shifts). A long-term driver like continental drift changes which species exist at all (new species evolve in isolation; old lineages go extinct). If a question says "explain over different time scales," you must contrast these two effects, not just list events.

The five drivers, up close

Each card pairs the driver with a real Australian example you can quote in an answer.

Wipe an area to bare rock, then trigger primary succession — new, fertile, isolated habitat that can later host unique species.

Swings rainfall. Droughts bleach reefs and shrink populations; La Niña floods trigger boom breeding in arid Australia.

Drifts continents into isolation, builds mountains and opens seaways — the master maker of endemism.

The engine. Natural selection fills new niches with new species (diversification) and lets ill-suited ones go extinct.

◆ NOTE

Volcanic eruptions — destroy, then create An eruption is the ultimate reset button: lava and ash sterilise the land, causing local extinction. But the cooled rock is a blank canvas. Primary succession — lichens → mosses → grasses → shrubs → forest — slowly rebuilds an ecosystem, and because the new ground is isolated, it can evolve its own species over time. Classic global cases: Krakatau (1883) and the island of Surtsey (1963), recolonised from zero. Right here in western Victoria, the Newer Volcanics Province (Tower Hill, Budj Bim / Mt Eccles) left ~400 vents and rich volcanic soils that shaped the region's wetlands and woodlands.

◆ NOTE

Fire — diversity's double-edged sword Australian plants are fire-adapted: eucalypts resprout from epicormic buds and lignotubers; many Banksia hold seed in woody cones and only release it after fire (serotiny); some seeds germinate only when triggered by smoke. A mild, patchy fire regime maintains diversity by clearing dominants and opening niches. But fires that are too frequent, too hot or too rare push species out. Fire-driven fragmentation on ancient soils is a key reason the South-West Australian Floristic Region radiated into thousands of endemic species.

◆ NOTE

El Niño & La Niña — boom and bust on repeat The El Niño–Southern Oscillation flips eastern Australia between dry and wet every few years. El Niño brings drought and heat — raising bushfire risk and driving mass coral bleaching on the Great Barrier Reef (2016, 2017, 2020). La Niña brings rain — and the arid interior booms: budgerigars and waterbirds breed in huge numbers. These swings drive boom–bust population cycles, and species adapted to ride them can become locally specialised over many cycles.

◆ NOTE

Plate tectonics — why Australia is so weird Australia broke from Gondwana and has drifted in near-isolation for ~45 million years. Isolation means species evolve with no gene flow from elsewhere, producing extraordinary endemism: around 80% of mammals, ~85% of flowering plants and ~90% of reptiles and frogs are found nowhere else. Tectonics also builds mountains (new alpine habitats), opens and closes seaways, and underlies biogeographic boundaries like Wallace's Line between Asian and Australian faunas.

◆ NOTE

Evolution & natural selection — the common engine Every other driver works through evolution. Isolation, fire, drought and fresh volcanic ground are all selection pressures. Over generations, natural selection adapts populations, and when populations stay isolated long enough they become new species (speciation / diversification). The same engine removes lineages that can't keep up (extinction). Evolution is what turns a geological accident into a unique living community.

Three words the drivers produce

Endemism, hotspots and extinction aren't separate topics — they're the outputs of the drivers above.

A species found in one region and nowhere else. Driven by isolation from plate tectonics, islands and mountains. Australian examples →

A region with exceptional endemism that is also highly threatened. SW Australia →

Loss of a species forever. Natural drivers cause it (volcanic winters, ENSO extremes, sea-level change) — but also clear niches that fuel new diversification.

◆ ANALOGY

Analogy · the island workshop Isolation is like locking a workshop for 45 million years. Cut off from outside designs, the tools inside get re-invented from scratch — you end up with gadgets (platypus, koala, banksia) that exist only in that room. That's endemism. Cram the most one-of-a-kind gadgets into one small, leaky room under threat, and you have a hotspot. Let the room flood once too often and some designs are lost forever — extinction.

Deep-time explorer

The five drivers sit on a logarithmic time axis from a few years to hundreds of millions of years. The markers nearest you act fastest.

Disturbance & recovery

Hit the ecosystem with a natural driver and watch diversity crash — then recover through succession. Fire and eruption behave differently. That difference is the point.

◆ NOTE

Read the curve like an examiner A bushfire in a fire-adapted system causes a sharp dip then fast recovery — seed banks, lignotubers and resprouting mean species bounce back (secondary succession). A volcanic eruption strips the land to bare rock: diversity falls near zero and rebuilds slowly from pioneers (primary succession). Same idea, very different time scale — exactly the contrast a "different time scales" question wants.

Decode the command word

◆ COMMAND WORDS

Half of losing marks is answering the wrong verb. Tap a card to see what the examiner actually wants for this topic.

Command word	What the marker wants
Identify / State	Name it — no explanation.
Describe	Give the features / what happens, in order.
Explain	Give the how / why — cause and effect.
Distinguish / Compare	State the difference (and similarity) explicitly.
Analyse	Break into parts and show how they relate.
Evaluate / Justify	Weigh strengths & limits, then a reasoned verdict.

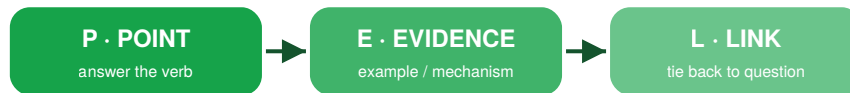
Match your answer to the verb — the fastest marks in the exam.

Exam traps to dodge

⚠ EXAM TRAPS

The mistakes that quietly cost marks on this dot point.

Writing trainer · P·E·L



The P·E·L answer shape — make a Point, support with Evidence, then Link to the question.

🔗 P·E·L WRITING

A reliable structure for "explain / analyse" answers: Point → Evidence → Link. Open each step to build a model paragraph.

Good answer vs bad answer

Same question, two responses. The highlights show exactly where marks are won and lost.

Practice questions

★ PRACTICE & SELF-MARK

Write your answer first, then reveal the model, tick off the marking points, and self-mark. Your total tracks in the score dock.

60 -second recap

- Five natural drivers: volcanic eruptions, fire, El Niño droughts/floods, plate tectonics, and evolution.
- Different time scales: fire & ENSO act over years–decades (which species are present); plate tectonics & evolution act over millions of years (which species exist).
- Endemism is mainly built by isolation — and Australia's drift from Gondwana makes it a world leader in endemic species.
- A hotspot = high endemism + high threat (e.g. the South-West Australian Floristic Region).
- Drivers cause extinction and diversification — disturbance clears niches that new species fill, all through evolution.

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Worked good/weak answers illustrate exam technique — always check against current VCAA marking advice.